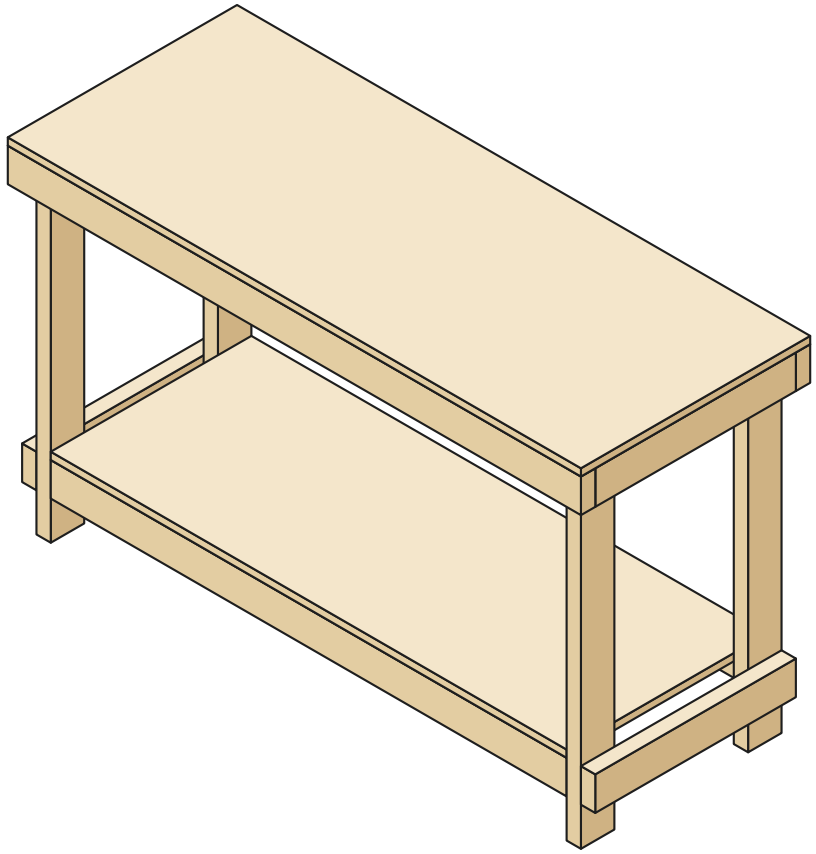
Garage Workbench

A sturdy 5 ft workbench from 2x4s with a 3/4 in plywood top and a lower shelf.



Difficulty	Time	Parts	Screws
Easy	**Half a day**	**15**	**103**

Shopping list

- 2x4 stud, pine/SPF, 1 1/2×3 1/2×144″ — 3 pcs
- 2x4 stud, pine/SPF, 1 1/2×3 1/2×96″ — 1 pcs
- 3/4 in plywood, 3/4×48×96″ — 1 pcs
- Wood screws #6 × 1 3/4″ — 59 pcs (+ a couple spare)
- Wood screws #10 × 2 3/4″ — 16 pcs (+ a couple spare)
- Wood screws #10 × 3 1/2″ — 28 pcs (+ a couple spare)
- Wood glue (optional)

Tools

- Circular saw or miter saw
- Drill/driver
- 1/8 in drill bit and countersink
- Tape measure, pencil, speed square
- Two clamps

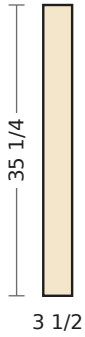
These instructions were generated automatically from a 3D model. Check the dimensions against your material before cutting. Structural strength is not calculated: you are responsible for safe building and use.

Parts

All dimensions in inches. Part letters match the letters in the assembly steps. Number pairs “a / b” are measured from the left and from the bottom edge of the part.

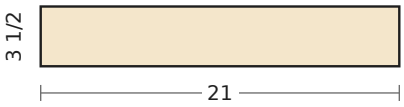
A Leg ×4

thickness 1 1/2″



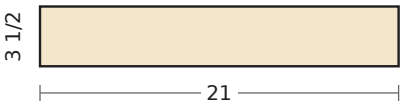
B Top end rail ×2

thickness 1 1/2″



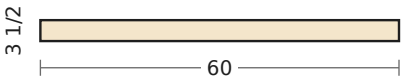
C Bottom end rail ×2

thickness 1 1/2″



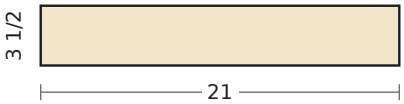
D Long apron ×2

thickness 1 1/2″



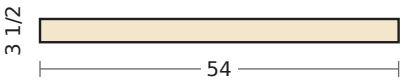
E Middle top rail ×1

thickness 1 1/2″



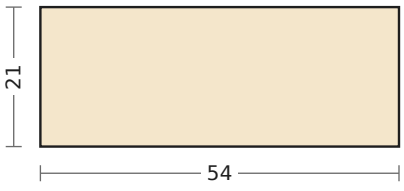
F Shelf rail ×2

thickness 1 1/2″



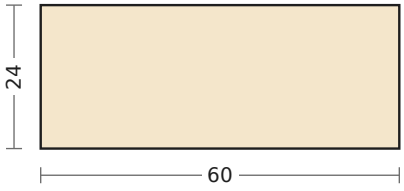
G Lower shelf ×1

thickness 3/4″



H Top ×1

thickness 3/4″



Cutting plan

Kerf of 1/8″ included. Crosscut the parts from the board. Easiest is to have the store cut the sheet — show them this layout.

2x4 stud, pine/SPF 1 1/2×3 1/2×144″ — #1



2x4 stud, pine/SPF 1 1/2×3 1/2×144″ — #2



2x4 stud, pine/SPF 1 1/2×3 1/2×144″ — #3



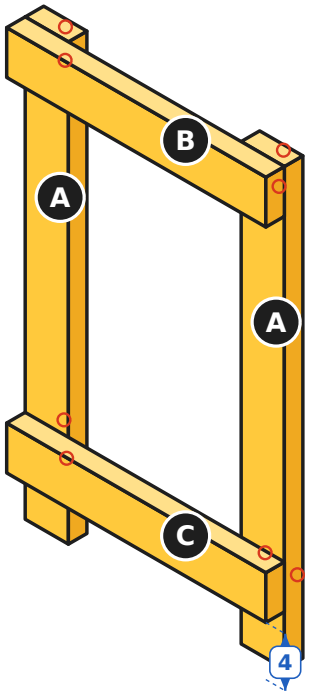
2x4 stud, pine/SPF 1 1/2×3 1/2×96″ — #4



3/4 in plywood 3/4×48×96″ — #1



1 Left end frame



Sub-assembly — build it separately on the bench

A **×2 Leg**

B **×1 Top end rail**

C **×1 Bottom end rail**

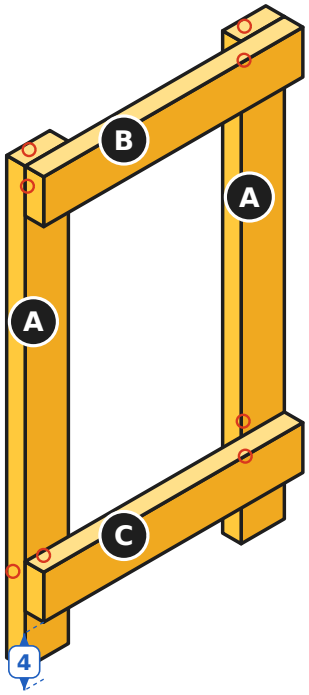
×8
screw #10 × 2 3/4″

↔ C: 4″ up from the bottom of A

○ — driven from the back side

Lay both legs flat on the floor, then screw the two end rails across them. Check the frame is square by measuring both diagonals - they should match.

2 Right end frame



Sub-assembly — build it separately on the bench

Same as in step 1, but mirrored

A **×2 Leg**

B **×1 Top end rail**

C **×1 Bottom end rail**

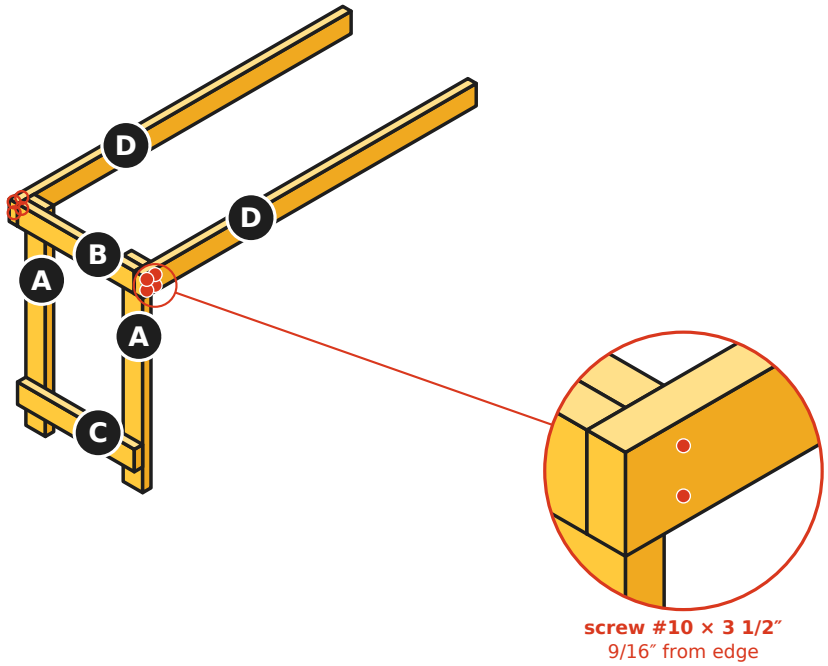
×8
screw #10 × 2 3/4″

↔ C: 4″ up from the bottom of A

○ — driven from the back side

Build the second frame as a mirror image: the rails go on the outside.

3 Long aprons



screw #10 × 3 1/2″
9/16″ from edge

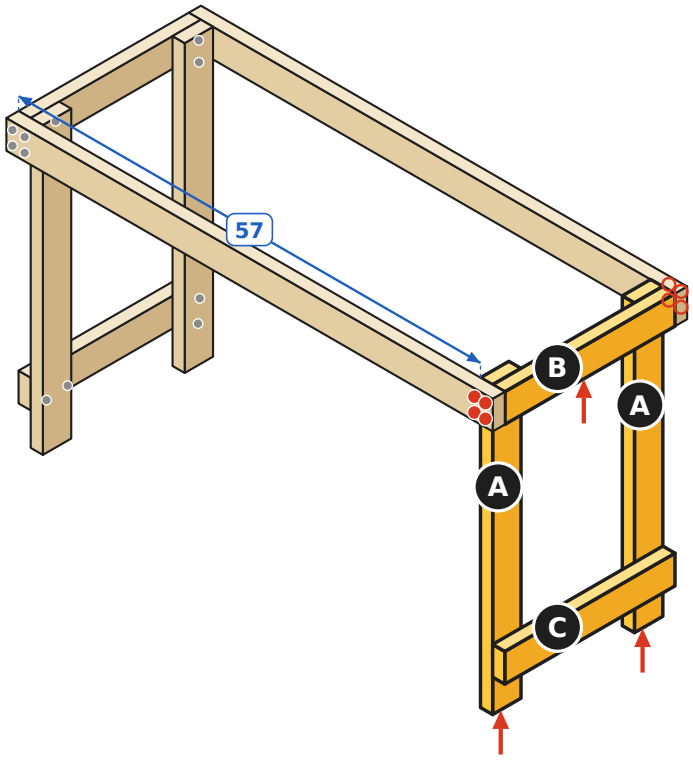
D ×2 Long apron

×8
screw #10 × 3 1/2″

● — screw positions; ○ — driven from the back side

Stand the left frame upright with the rails facing out. Screw the front and back aprons to the left frame, flush with the top of the legs. A helper or clamps make this much easier.

4 Attach the right frame



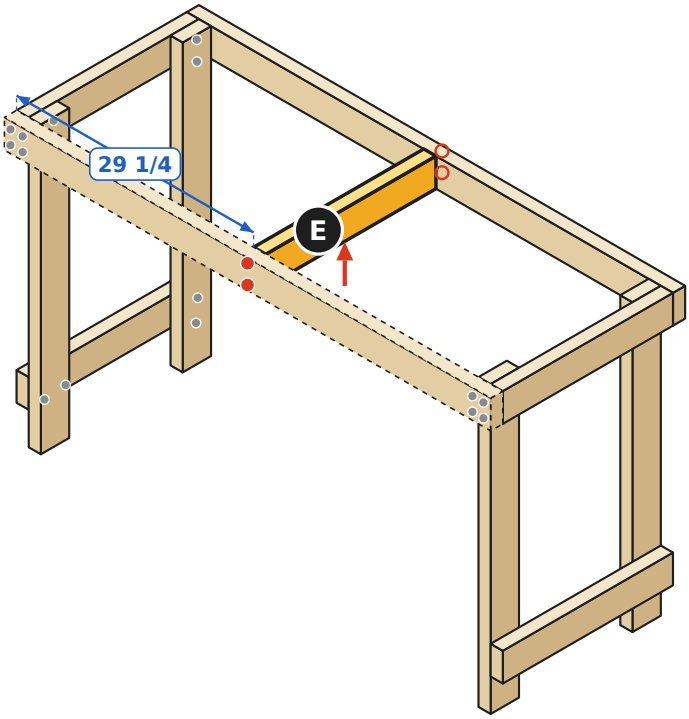
×8
screw #10 × 3 1/2″

↔ A: 57″ in from the left edge of D

● — screw positions; ○ — driven from the back side

Slide the right frame between the aprons, flush at the top, and screw the aprons to it.

5 Middle rail



E **×1 Middle top rail**

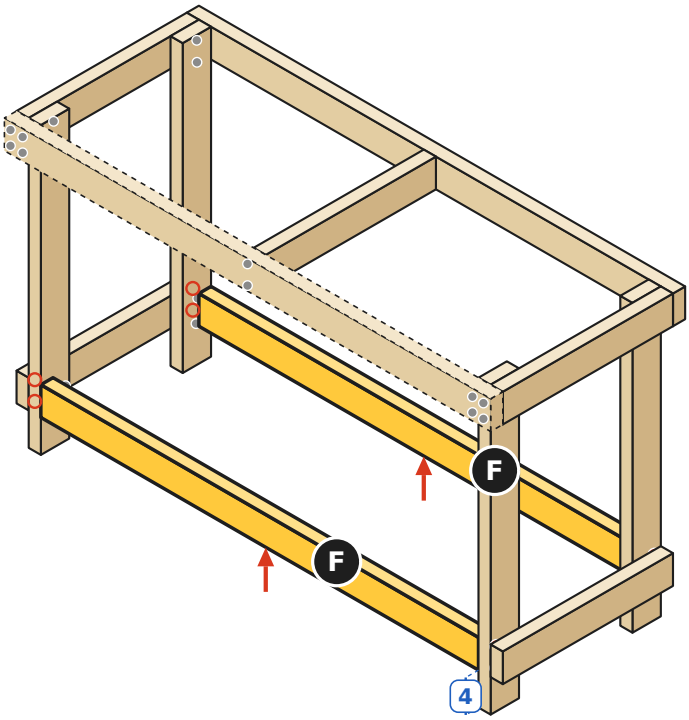
×4
screw #10 × 3 1/2″

↔ E: 29 1/4″ in from the left edge of D

● — screw positions; ○ — driven from the back side

Fit the middle rail between the aprons, flush with their top. It supports the plywood in the middle.

6 Shelf rails



F **×2 Shelf rail**

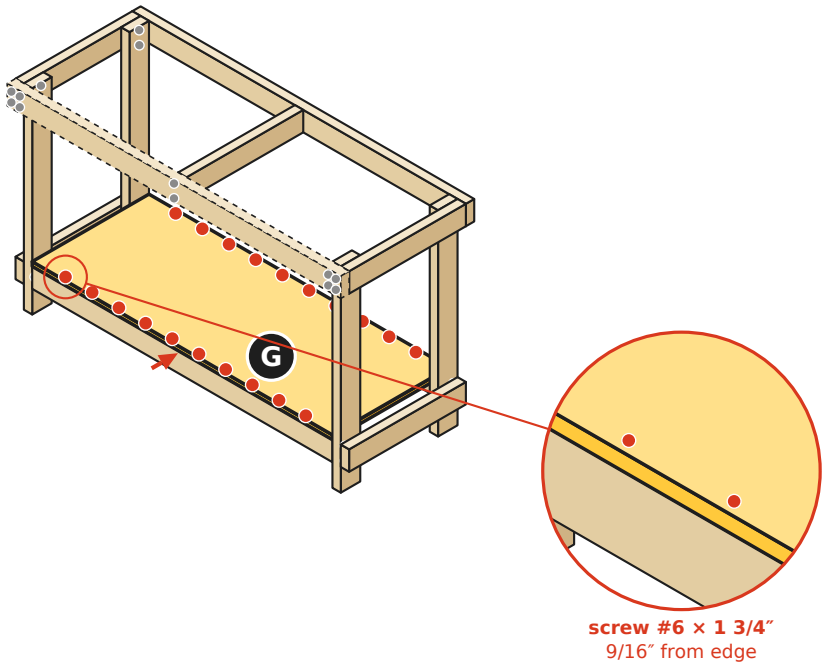
×8
screw #10 × 3 1/2″

↔ F: 4″ up from the bottom of A

● — screw positions; ○ — driven from the back side

Fit the two shelf rails between the legs, level with the bottom end rails, and screw through the legs.

7 Lower shelf



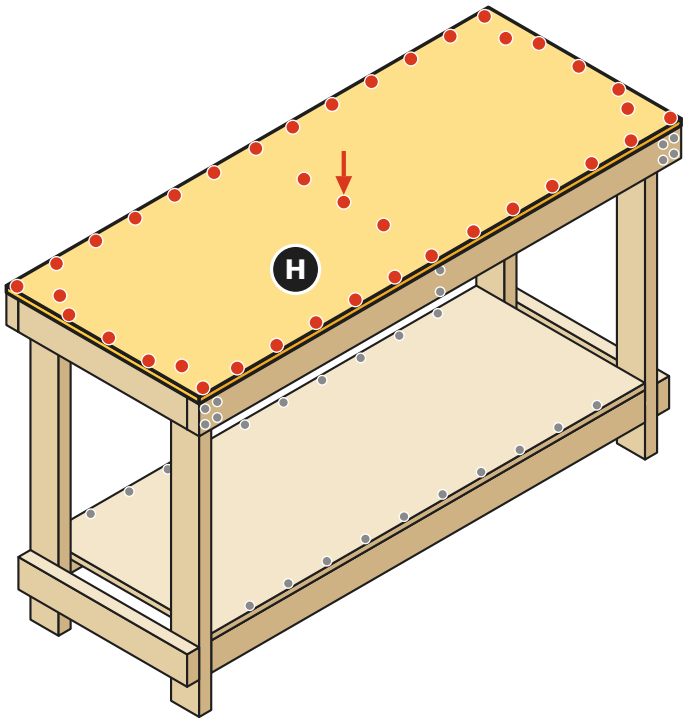
G ×1 Lower shelf

×20
screw #6 × 1 3/4″

● — screw positions

Drop the shelf onto the rails and screw it down to them.

8 Top



H ×1 Top

×39
screw #6 × 1 3/4″

● — screw positions

Lay the plywood top flush with the frame on all sides. Screw it into the aprons, end rails and middle rail every 8-10 inches.

Important

- Cut all four legs together, clamped side by side, so they come out exactly the same length.
- If the garage floor is uneven, shim under a leg rather than cutting it.

Safety

- Load-bearing build: strength is only roughly estimated from shelf sag. Do not load it beyond the figures below; when in doubt, add fasteners or use thicker stock.
- Wear safety glasses and follow the manufacturer's instructions for power tools. Keep fingers away from the cut line.

Design loads

• Lower shelf (G): 50 kg spread evenly — sag about 0″ over a 19 1/2″ span
• Top (H): 100 kg spread evenly — sag about 0″ over a 22 1/2″ span
• Stability: tips over when leaned about 22°.

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